

SCIE1500: Analytical Methods for Scientists

SAMPLE FINAL EXAM 2

The University of Western Australia — Semester 2, 2026

Purpose of this document. This is a *second* sample exam, provided in response to student requests for more practice. It follows exactly the same three-part structure, mark allocation and timing as the actual exam, and as Sample Exam 1. None of the questions below appears in the real paper, and none repeats a question from Sample Exam 1.

Exam format. Three parts, 110 marks, 2 hours:

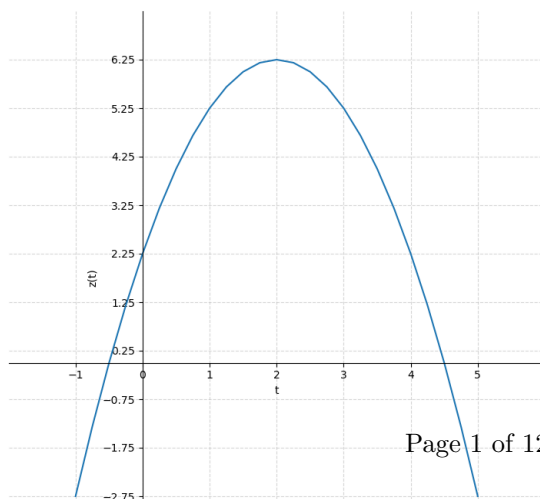
- **Part I — Multiple Choice** (72 marks; 2 marks each): 36 questions spanning all twelve weeks. Answer on the MCQ answer sheet.
- **Part II — Python as a Mathematical Tool** (14 marks; 2 marks each): 7 multiple-choice questions. *You will not need a computer.* Answer on the MCQ answer sheet.
- **Part III — Long Answer** (24 marks; 12 marks per question): **Answer any 2 of the 4** questions provided. Show all working in the answer booklet.

A note on the source of these questions. Some Part I questions are drawn from earlier SCIE1500 papers. Those papers had a different structure — two parts, no Python section, and all four long-answer questions compulsory — so do *not* use the original papers as a guide to the format. This document, not the older paper, reflects what you will sit.

PART I — Multiple Choice Questions (72 marks; 2 marks per question)

Answer ALL multiple choice questions using the MCQ answer sheet.

1. In which of the following cases is the Domain (D) for the function not identified correctly?
 - (a) $y = -3 + 6x$, $D = \{x : x \in \mathbb{R}\}$.
 - (b) $y = \frac{(2x-1)^{\frac{1}{2}}}{2} - 2$, $D = \{x : x \in \mathbb{R}\}$.
 - (c) $y = \frac{2}{1-e^{-x}}$, $D = \{x \in \mathbb{R} : x \neq 0\}$.
 - (d) $y = e^{-0.5x+2}$, $D = \{x : x \in \mathbb{R}\}$.
 - (e) None of the above.
2. Which of the following quadratic functions does the plot shown below correspond to?



- (a) $y = -4x^2 + x + \frac{81}{4}$ (d) $y = -x^2 + 4x + \frac{9}{4}$
 (b) $y = -x^2 + 5x + \frac{81}{4}$
 (c) $y = -4x^2 + x + \frac{9}{4}$ (e) $y = x^2 - 4x + \frac{9}{4}$
3. If $f(x) = -e^{-2x+1}$ and $g(x) = -\frac{1}{2}x^2$, what is $f \circ g(x)$?
 (a) $(-e^{-2x+1})(-\frac{1}{2})$
 (b) $-e^{x^2+1}$
 (c) $\frac{1}{2}e^{4x^2-4x+1}$
 (d) xe^{-2x+1}
 (e) None of the above.
4. Which of the following is not true about the function $y = (x + 2)(x - 3)$?
 (a) Its range is $\{y \in \mathbb{R} : y \geq -6.25\}$.
 (b) It is a concave up function, i.e. it opens up.
 (c) The slope of its tangent at the point where x is 2 is 3.
 (d) Its second derivative is $y'' = 2$.
 (e) None of the above.
5. A storage tank is draining. The volume remaining after t minutes is $V(t) = 600 - 30t$ litres. Which of the following correctly identifies the *mathematical* and *physical* domains of V ?
 (a) Mathematical domain $\mathbb{R}$; physical domain $[0, 20]$.
 (b) Mathematical domain $[0, 20]$; physical domain $\mathbb{R}$.
 (c) Both domains are equal to $\mathbb{R}$.
 (d) Mathematical domain $\mathbb{R}$; physical domain $[0, 600]$.
 (e) None of the above.
6. Consider a population that grows at the rate of 5 percent per week (i.e. $r_m = 0.05$). If the longevity of individuals in this species is 22 weeks on average, what is the R_o factor for this population? Use the discrete-generation approximation $R_0 \approx 1 + r_m L$, where L is reproductive longevity. R_o is the rate of increase per head per generation. Provide an answer that is correct when rounded to a whole number.
 (a) 12
 (b) 120
 (c) 2
 (d) 24
 (e) None of the above.
7. Suppose the probability (p) that a person contracts a disease is described as function of risk (x):

$$p = \frac{1}{1+e^{-3x}}$$

Which of the following statements is not correct?

- (a) p is $\frac{1}{2}$ if $x = 0$.
 (b) p is below $\frac{1}{2}$ if x is negative.
 (c) p is an increasing function of x .
 (d) p approaches zero if x goes towards $-\infty$.
 (e) None of the above.
8. The concentration of a drug in the bloodstream decays according to $C(t) = 80e^{-0.25t}$ mg/L, where t is in hours. After how many hours does the concentration fall to 10 mg/L?

- (a) 2.77 hours
 (b) 5.55 hours
 (c) 8.32 hours
 (d) 32 hours
 (e) None of the above.
9. Which of the following is equal to $\ln\left(\frac{x^3\sqrt{y}}{z}\right)$ for $x, y, z > 0$?
- (a) $3\ln x + 2\ln y - \ln z$
 (b) $3\ln x + \frac{1}{2}\ln y - \ln z$
 (c) $\ln(3x) + \ln(\sqrt{y}) - \ln z$
 (d) $\frac{3\ln x \cdot \frac{1}{2}\ln y}{\ln z}$
 (e) None of the above.
10. Consider a population trajectory over time (t) which can be described by the logistic equation: $P(t) = \frac{K}{1+ Ae^{-\alpha t}}$. The values of the parameters in the model are as follows: $K = 15000$, $A = 120$ and $\alpha = 0.15$. How many time periods would it take for this population to reach 50% of its carrying capacity? (Round your answer to a whole number.)
- (a) 32 time periods
 (b) 31 time periods
 (c) 27 time periods
 (d) 50 time periods
 (e) None of the above
11. Consider the Schaefer (1957) model of growth:
- $$G(S) = g \cdot S \cdot \left(1 - \frac{S}{K}\right)$$
- where $G(S)$ is fish growth, g is the intrinsic growth rate, S is the stock level and K is the carrying capacity. Which of the following statements is true?
- (a) The actual fish growth rate, defined as the ratio of $G(S)$ to S , declines as stock increases.
 (b) Fish growth (or $G(S)$) is highest when the stock (S) is half the maximum carrying capacity (K).
 (c) Fish growth (or $G(S)$) is always non-negative.
 (d) All of the above.
 (e) Only (b) and (c) are correct.
12. What is the derivative of the function $y = \frac{3}{4}x^4 + \frac{1}{4}x^2 - 3x + \ln x$?
- (a) $y' = 12x^3 + \frac{3}{x^2} - \frac{1}{2x^{\frac{3}{2}}}$
 (b) $y' = x^3 - 3 + \frac{1}{2}x$
 (c) $y' = \frac{3}{5}x^5 - \frac{3}{2}x^2 - \frac{3}{2}x$
 (d) $y' = 3x^3 + \frac{1}{2}x - 3 + \frac{1}{x}$
 (e) None of the above.
13. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.
- (a) 0
 (b) 3
 (c) 6
 (d) The limit does not exist, because the denominator is zero at $x = 3$.

- (e) None of the above.
14. For which values of x does the curve $y = x^3 - 12x + 5$ have a *horizontal* tangent?
- $x = 0$ only
 - $x = 4$ and $x = -4$
 - $x = 2$ only
 - $x = 2$ and $x = -2$
 - None of the above.
15. If $y = \frac{2x+1}{x^2+3}$, then $\frac{dy}{dx}$ equals
- $\frac{2}{2x}$
 - $\frac{(2x+1)(2x) - 2(x^2+3)}{(x^2+3)^2}$
 - $\frac{2}{(x^2+3)^2}$
 - $\frac{2(x^2+3) - (2x+1)(2x)}{x^2+3}$
 - None of the above.
16. A rectangular enclosure is to be built against a straight river bank, so fencing is needed on only three sides. With 120 m of fencing available, let x be the length of each side perpendicular to the river and y the side parallel to it. Which dimensions maximise the enclosed area?
- $x = 30$ m, $y = 60$ m, area 1800 m²
 - $x = 40$ m, $y = 40$ m, area 1600 m²
 - $x = 20$ m, $y = 80$ m, area 1600 m²
 - $x = 60$ m, $y = 0$ m, area 0 m²
 - None of the above.
17. If $y = (3x^2 + 1)^5$, then $\frac{dy}{dx}$ equals
- $5(3x^2 + 1)^4$
 - $30x(3x^2 + 1)^4$
 - $6x(3x^2 + 1)^5$
 - $15x(3x^2 + 1)^4$
 - None of the above.
18. The indefinite integral $\int \left(\frac{3}{x^2} + 4\sqrt{x} \right) dx$ equals
- $-\frac{3}{x} + \frac{8}{3}x^{3/2} + C$
 - $-\frac{3}{x} + \frac{4}{3}x^{3/2} + C$
 - $\frac{3}{x} + \frac{8}{3}x^{3/2} + C$
 - $-6x^{-3} + 2x^{-1/2} + C$
 - None of the above.
19. The indefinite integral $\int \left(\frac{1}{x} + e^{2x} \right) dx$ equals
- $\ln|x| + 2e^{2x} + C$
 - $-\frac{1}{x^2} + e^{2x} + C$

- (c) $\ln|x| + \frac{1}{2}e^{2x} + C$
 (d) $\ln|x| + e^{2x} + C$
 (e) None of the above.
20. Suppose $F'(x) = 6x^2 - 4x$ and $F(1) = 5$. Then $F(x)$ equals
 (a) $2x^3 - 2x^2 + 5$
 (b) $2x^3 - 2x^2 + 1$
 (c) $12x - 4 + 5$
 (d) $2x^3 - 2x^2$
 (e) None of the above.
21. Let f be continuous on $[a, b]$, let $\Delta x = (b - a)/n$ and let $x_i = a + i \Delta x$. Which of the following statements is *correct*?
 (a) $\sum_{i=1}^n f(x_i) \Delta x$ equals $\int_a^b f(x) dx$ exactly, for every n .
 (b) $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \int_a^b f(x) dx$.
 (c) The sum $\sum_{i=1}^n f(x_i) \Delta x$ approximates the derivative of f on $[a, b]$.
 (d) $\sum_{i=1}^n f(x_i) \Delta x$ equals $f(b) - f(a)$.
 (e) None of the above.
22. Consider the sequence $a_i = 3 + 5(i - 1)$ for $i = 1, 2, 3, \dots$. How many of the terms in this sequence are greater or equal to 10 and less than or equal to 150?
 (a) 27
 (b) 28
 (c) 29
 (d) 30
 (e) None of the above.
23. What is a geometric sequence?
 (a) A sequence of numbers with a common ratio between two consecutive numbers.
 (b) A sequence of numbers with a constant consecutive difference between them.
 (c) Any sequence of numbers, as long as a number is a product of the preceding number.
 (d) A sequence of numbers where the ratio between the $i - th$ and the $(i + 1) - th$ term is equal to the common ratio raised to the power of i .
 (e) A sequence of numbers with a common ratio that is between 0 and 1.
24. Consider the following Lotka-Volterra model, where the prey population is denoted by H and the predator population by P .

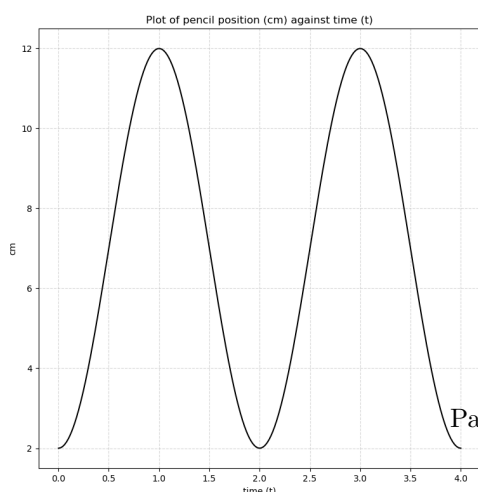
$$\begin{aligned}\frac{dH}{dt} &= \alpha H - \beta HP \\ \frac{dP}{dt} &= \lambda HP - \gamma P\end{aligned}$$

The parameter values are $\alpha = 0.15, \beta = 0.005, \lambda = 0.0005$, & $\gamma = 0.10$. Which of the following is not true?

- (a) $H = 0, P = 0$ is a fixed point for this system.
 (b) $H = 200, P = 30$ is a fixed point for this system.
 (c) The efficiency of predation is 10%.
 (d) The efficiency of predation is 0.05%.
 (e) None of the above.

25. Consider a Lotka-Volterra predator–prey system with a non-trivial equilibrium (H^*, P^*) , where H is prey density and P is predator density. Which of the following statements is *correct*?
- Trajectories spiral inward and converge to (H^*, P^*) .
 - Both populations grow without bound once they leave the equilibrium.
 - The populations cycle on closed orbits around (H^*, P^*) , with peaks in predator numbers lagging behind peaks in prey numbers.
 - The equilibrium is a saddle point, so almost all trajectories diverge from it.
 - None of the above.
26. If five coins are tossed, which of the following is true?
- The size of the sample space is 10, i.e. 5×2 .
 - The size of the sample space is 32, i.e. 2^5 .
 - The probability of getting exactly three heads is 0.60.
 - The probability of getting zero heads is 0.375.
 - None of the above.
27. In a cohort of 200 students, 80 study biology, 60 study chemistry, and 25 study both. A student is chosen at random and turns out to study biology. What is the probability that this student also studies chemistry?
- 0.125
 - 0.417
 - 0.575
 - 0.200
 - None of the above.
28. A box contains 5 red and 7 blue marbles. Two marbles are drawn *without* replacement. What is the probability that both are red?
- $\frac{25}{144}$
 - $\frac{5}{12}$
 - $\frac{5}{33}$
 - $\frac{1}{6}$
 - None of the above.
29. In how many different ways can a committee of 3 people be chosen from a group of 8?
- 24
 - 56
 - 336
 - 512
 - None of the above.
30. What is the expected value of the random variable X with the probabilities of occurrence $P(X)$ shown below?
- | | | |
|-----|--------|------------------------|
| X | $P(X)$ | (a) 1.0 |
| 0 | 0.2 | (b) 2.0 |
| 3 | 0.4 | (c) 3.0 |
| 4 | 0.3 | (d) 4.0 |
| 6 | 0.1 | (e) None of the above. |

31. Health experts suspect that a new variant of a virus is more infectious than an older one, which had an infection rate (p) of 50%, i.e. one out of two non-infected persons coming in contact with an infected person would be expected to contract the disease. Suppose you examined a sample of contact tracing data and found that out of 11 recent contact cases, 9 resulted in infections. And suppose you used your knowledge of binomial distributions to determine that the probability of observing 9 or more infections out of 11 contact cases would be $\frac{67}{2^{11}}$, or about 0.033, if the true infection rate (p) is indeed 0.50. If you wanted to do a hypothesis test to determine if there is empirical support for the claim that the new variant is more infectious, how would you proceed?
- Start with the null and alternative hypotheses set as follows: $H_o : p \leq 0.50$, $H_a : p > 0.50$; then draw the conclusion that the evidence rejects the null at the 5% level of statistical significance.
 - Start with the null and alternative hypotheses set as follows: $H_o : p = 0.50$, $H_a : p \neq 0.50$; then draw the conclusion that the evidence rejects the null at the 5% level of statistical significance.
 - Start with the null and alternative hypotheses set as follows: $H_o : p \leq 0.50$, $H_a : p > 0.50$; then draw the conclusion that the evidence is not sufficient to reject the null at the 5% level of statistical significance.
 - Start with the null and alternative hypotheses set as follows: $H_o : p = 0.50$, $H_a : p \neq 0.50$; then draw the conclusion that the evidence is not sufficient to reject the null at the 5% level of statistical significance.
 - Start with the null and alternative hypotheses set as follows: $H_o : p \geq 0.50$, $H_a : p < 0.50$; then draw the conclusion that the evidence is not sufficient to reject the null at the 5% level of statistical significance.
32. A discrete random variable X takes the values 1, 2 and 3 with probabilities 0.2, 0.5 and 0.3 respectively. What is the *variance* of X ?
- 2.1
 - 0.7
 - 4.9
 - 0.49
 - None of the above.
33. Which of the following statements is true?
- 90° converted to radians is equal to $\frac{1}{2}\pi$.
 - The function $\frac{5}{2}\sin(2x)$ has an amplitude of 2.5.
 - The period of the function $x(t) = \frac{7}{2}\cos(\pi t)$ is 2 seconds if time (t) is measured in seconds.
 - The frequency of the function $x(t) = \frac{7}{2}\cos(\pi t)$ is 0.5Hz if time (t) is measured in seconds.
 - All of the above are true.
34. In a research experiment on human motor behaviour, a person moves a pencil on a sheet of paper periodically between the positions 2cm and 12cm completing a round every 2 seconds. See plot of pencil position against time below on the left. If you make a model of the pencil's location at time t , $x(t) = A \cos(B(t + C)) + D$, what should be the values of A , B , C and D if it is known that the pencil's tip was at position $x = 2\text{cm}$ at time $t = 0$?



- (a) $A = 6, B = \pi, C = 1, D = 7$
- (b) $A = 6, B = \pi, C = 1.5, D = 8$
- (c) $A = 5, B = 2\pi, C = 1.5, D = 7$
- (d) $A = 5, B = \pi, C = 1, D = 7$
- (e) $A = 12, B = 2\pi, C = -1.5, D = 8$

35. The demand and supply equations for a commodity are $Q^d = 120 - 3P$ and $Q^s = 20 + 2P$, where P is the price. What are the equilibrium price and quantity?
- (a) $P = 25, Q = 45$
 - (b) $P = 20, Q = 45$
 - (c) $P = 15, Q = 75$
 - (d) $P = 20, Q = 60$
 - (e) None of the above.
36. A firm maximises $Z = 5x + 4y$ subject to $x + y \leq 40$, $2x + y \leq 60$ and $x, y \geq 0$. The corner points of the feasible region are $(0, 0)$, $(30, 0)$, $(20, 20)$ and $(0, 40)$. By the corner-point theorem, which point gives the maximum, and what is Z there?
- (a) $(30, 0)$ with $Z = 150$
 - (b) $(0, 40)$ with $Z = 160$
 - (c) $(0, 0)$ with $Z = 0$
 - (d) $(20, 20)$ with $Z = 160$
 - (e) None of the above.

PART II — Python as a Mathematical Tool (14 marks; 2 marks per question)

Answer ALL questions using the MCQ answer sheet. No computer is required. All questions are answerable from your understanding of how the Python tools used in the lab notebooks work.

37. Consider the following code:

```
import numpy as np
x = np.linspace(0, 10, 5)
print(x)
```

What is printed?

- (a) [0 1 2 3 4 5 6 7 8 9 10] — every integer from 0 to 10.
- (b) [0. 2.5 5. 7.5 10.] — five evenly spaced values including both endpoints.
- (c) [0 2 4 6 8] — five values in steps of 2, stopping before 10.
- (d) Five random values drawn between 0 and 10.
- (e) None of the above.

38. A student examines the long-run behaviour of a population model:

```
import sympy as sp
t = sp.symbols('t')
P = 5000*t / (t + 20)
L = sp.limit(P, t, sp.oo)
```

What does L contain?

- (a) 5000 — the limiting population as $t \rightarrow \infty$.
- (b) 0, because the denominator grows without bound.
- (c) 250, the value of P at $t = 20$.
- (d) The derivative dP/dt evaluated at large t .
- (e) None of the above.

39. A student solves for a market equilibrium:

```
import sympy as sp
P = sp.symbols('P', positive=True)
demand = 120 - 3*P
supply = 20 + 2*P
result = sp.solve(sp.Eq(demand, supply), P)
```

What does `result` contain, and what does it mean?

- (a) [60] — the equilibrium quantity traded.
- (b) [100] — the excess demand at a price of zero.
- (c) [20] — the equilibrium price, at which quantity demanded equals quantity supplied.
- (d) [120, 20] — the intercepts of the two curves.
- (e) None of the above.

40. A student simulates a predator–prey system:

```
import numpy as np
from scipy.integrate import odeint

def model(y, t):
    H, P = y
    return [0.10*H - 0.004*H*P,
            0.0004*H*P - 0.08*P]
```

```
t = np.linspace(0, 200, 2000)
sol = odeint(model, [300, 15], t)
```

What does `sol` contain?

- (a) The two equilibrium values H^* and P^* .
- (b) A 2000×2 array giving the prey and predator densities at each of the 2000 time points.
- (c) The derivatives dH/dt and dP/dt evaluated at $t = 0$ only.
- (d) A symbolic formula for $H(t)$ and $P(t)$.
- (e) None of the above.

41. Consider the following code:

```
from math import comb
ways = comb(8, 3)
```

What does `ways` equal, and what does it count?

- (a) 336 — the number of *ordered* arrangements of 3 people drawn from 8.
- (b) 56 — the number of distinct committees of 3 that can be formed from 8 people.
- (c) 24 — the product 8×3 .
- (d) 512 — the value 8^3 .
- (e) None of the above.

42. A student models a pendulum's displacement:

```
import numpy as np
t = np.linspace(0, 4, 400)
x = 3*np.cos(np.pi*t) + 7
```

What are the amplitude, period and vertical shift of `x`?

- (a) Amplitude 3, period 2 s, vertical shift 7.
- (b) Amplitude 3, period π s, vertical shift 7.
- (c) Amplitude 7, period 2 s, vertical shift 3.
- (d) Amplitude 3, period $\frac{1}{2}$ s, vertical shift 7.
- (e) None of the above.

43. A student shades the feasible region of a linear program:

```
import numpy as np
x = np.linspace(0, 40, 400)
y = np.linspace(0, 60, 400)
X, Y = np.meshgrid(x, y)
feasible = (X + Y <= 40) & (2*X + Y <= 60)
```

What does `feasible` contain?

- (a) The coordinates of the optimal corner point.
- (b) The value of the objective function at every grid point.
- (c) A 400×400 array of `True/False` values, marking which grid points satisfy *both* constraints.
- (d) A list of the four corner points of the feasible region.
- (e) None of the above.

PART III — Long-Answer Questions (24 marks; 12 marks per question)

Answer ANY TWO of the following four questions in the answer booklet. Show all working. You are required to answer only 2 of the 4 questions in this section.

44. (12 marks) Suppose the daytime variation in the number of white blood cells (lymphocytes) can be described as:

$$W'(t) = -0.15 + 0.091t - 0.009t^2 \text{ (Bn cells/L/hr)}$$

where Bn denotes billion, L litre and hr hour, and daytime is defined as between 8am ($t=0$) and 6pm ($t=10$), or the hours when clinics are open.

- (a) (3 points) Find the antiderivative of $W'(t)$.
 - (b) (3 points) Suppose the patient's blood cell count at 2pm is expected to be 2.39 billion. Write the equation ($W(t)$) for the number of lymphocytes per litre at time t .
 - (c) (3 points) Use $W(t)$ to find the expected lymphocyte count at 8am ($t = 0$) and at 6pm ($t = 10$).
 - (d) (3 points) Compute the *total change* in the lymphocyte count between 8am and 6pm, and explain what the sign of that change means for a clinic scheduling blood tests.
45. (12 marks) A farmer is fattening 100 deer for sale. The deer are currently gaining weight at the rate of 2kg per day, but this rate of gain declines by 100 grams each day into the future. The deer currently weigh 50kg each and cost \$0.50 per animal per day in terms of feed and other operating costs. Suppose you have been asked to advise the farmer on the best time to sell the deer to get the maximum possible profit, with profit defined as the difference between total revenue from the sale of the deer and the total cost incurred between now and the time of sale. The market price of deer is expected to be \$5.00 per kg for the foreseeable future.
- (a) (4 points) Show how you would set up a profit maximisation problem for this problem. Use t to denote number of days and π to denote profit.
 - (b) (5 points) Use calculus to solve the optimisation problem and determine for how many more days the farmer should keep the animals on the farm to maximise profit.
 - (c) (3 points) Present the optimal values of the following: total revenue from sale of deer, total cost of maintaining the deer, and the profit.
46. (12 marks) The demand and supply equations for a commodity are as shown below:

$$\text{Demand: } Q^d = 100 - 2P$$

$$\text{Supply: } Q^s = -100 + 3P$$

where P is the market price per unit and Q^d and Q^s are the quantity demanded and supplied, respectively. Answer the following questions.

- (a) (2 points) What is the maximum price any consumer would pay for this product?
 - (b) (2 points) What is the minimum price producers would consider supplying the product to the market at?
 - (c) (4 points) Find the market equilibrium price and quantity values.
 - (d) (2 points) What is the consumer surplus (CS) associated with the equilibrium outcomes?
 - (e) (2 points) What is the producer surplus (PS) associated with the equilibrium outcomes?
47. (12 marks) There are two food items (Food I and Food II) that can be combined to design a diet satisfying a set of minimum nutrient requirements. The nutrient content of the two food items as well as the minimum requirements are as shown below. For example, a unit of Food I offers 4 units of protein, while a unit of Food II offers 5. The diet created should have at least 100 units of protein, 400 units of energy, 10 units of Vitamin A and 5 units of calcium. The cost of food items I and II is \$2 and \$3, respectively. You have been asked to find the least costly diet that combines the two foods to provide at least the minimum dietary requirements.

Nutrients	Nutrient Content		Minimum Diet Requirements
	Food I	Food II	
Protein	4.00	5.00	100.00
Energy	10.00	25.00	400.00
Vitamin A	0.80	0.40	10.00
Calcium	0.00	1.00	5.00

- (a) (3 points) Formulate a linear programming (LP) problem for finding the cost minimising combination of the two food items. Use x and y to denote the number of food items I and II used to constitute the diet.
- (b) (4 points) Draw the feasible region for the LP problem on the graph paper provided to you. The feasible region is all the combinations of Food I and Food II that satisfy the dietary requirements. The drawing should show the feasible region as a shaded area. Label the vertices of the feasible region using letters A, B, ..., so that you can easily refer to them when answering the next two questions.
- (c) (3 points) How many units of Food I and Food II should be combined to make the desired diet in the least costly way?
- (d) (2 points) How would your answer to the question above change if the cost of Food I increased to \$5 per unit while that of Food II remained unchanged.

— END OF SAMPLE EXAM 2 —